

Vending Machine Controller^{1/7}

FPGA Implementation

Xilinx Spartan-7 Boolean Board

Project Overview

- FSM-based vending machine controller with integrated datapath
- Two-product selection system with real-time stock management
- Hardware debouncing for reliable button inputs (10ms threshold)
- Seven-segment display for change/return visualization (0-150 range)
- Complete error handling and transaction validation
- Minimal resource utilization (<1% LUTs, <1% FFs)

Team Members

Sohom Roy - 23EC8033

Digonto Paul - 23EC8053

Triparno Chakraborty - 23EC8051

Abir Dutta - 23EC8067

```
graph TD; TM[TOP MODULE (top_module)] --> D[DEBOUNCER (x4)  
coin_btn, pq_btn, buy_btn, go_btn]; D -- Clean Signals --> VM[VENDING MACHINE  
FSM + Datapath]; TM --> SS[SEVEN SEGMENT  
Display Controller]; VM --> SO[STATUS OUTPUTS  
done, error, LEDs];
```

The diagram illustrates the system architecture. At the top is the **TOP MODULE (top_module)**. It connects to a **DEBOUNCER (x4)** which outputs **Clean Signals** to the **VENDING MACHINE (FSM + Datapath)**. The **TOP MODULE** also connects to a **SEVEN SEGMENT Display Controller**. The **VENDING MACHINE** outputs **STATUS OUTPUTS** (done, error, LEDs).

- ▶ return_display → 7-seg
- ▶ balance_display (internal)
- ▶ done, error, LEDs
- ▶ selection/status signals

Arithmetic and Logic Units

Cost Calculation Unit

- ▶ **Operation:** $\text{total_cost} = (5 \times \text{quantity_p1}) + (10 \times \text{quantity_p2})$
- ▶ **Implementation:** Combinational multipliers
- ▶ **Product 1 Price:** ₹5 (PRICE_P1 parameter)
- ▶ **Product 2 Price:** ₹10 (PRICE_P2 parameter)
- ▶ **Output:** 8-bit total_cost wire

Validation Logic

- ▶ **sufficient_balance:** $(\text{balance_display} \geq \text{total_cost})$
- ▶ **sufficient_stock:** $(\text{stock_p1} \geq \text{qty_p1}) \text{ AND } (\text{stock_p2} \geq \text{qty_p2})$
- ▶ **Change Calculation:** $\text{return_display} = \text{balance} - \text{total_cost}$
- ▶ **Implementation:** Combinational comparators

Datapath Operations by State

IDLE State Operations:

- ▶ $\text{balance_display} \leftarrow 0$
- ▶ $\text{return_display} \leftarrow 0$
- ▶ $\text{quantity_p1} \leftarrow 0$
- ▶ $\text{quantity_p2} \leftarrow 0$

PURCHASE State Operations:

- ▶ coin: $\text{balance} \leftarrow \text{balance} + 5$
- ▶ $\text{pq} + \text{s1s2}=01$: $\text{qty_p1} \leftarrow \text{qty_p1} + 1$
- ▶ $\text{pq} + \text{s1s2}=10$: $\text{qty_p2} \leftarrow \text{qty_p2} + 1$
- ▶ Update status LEDs

DONE State Operations:

- ▶ $\text{stock_p1} \leftarrow \text{stock_p1} - \text{qty_p1}$
- ▶ $\text{stock_p2} \leftarrow \text{stock_p2} - \text{qty_p2}$
- ▶ $\text{return_display} \leftarrow \text{balance} - \text{total_cost}$
- ▶ $\text{balance_display} \leftarrow \text{balance} - \text{total_cost}$

ERROR State Operations:

- ▶ $\text{return_display} \leftarrow \text{balance}$ (full refund)
- ▶ No stock deduction
- ▶ Preserve balance for retry

Seven-Segment Display Datapath

BCD Conversion & Multiplexing

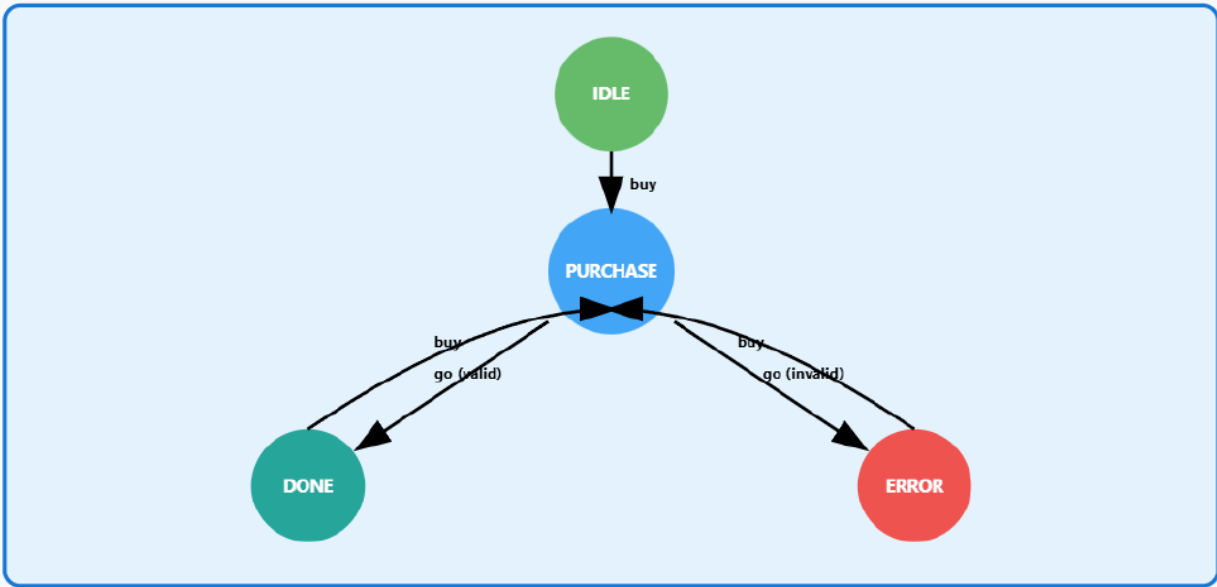
Binary-to-BCD Converter:

- ▶ **Algorithm:** Double-Dabble (Shift-Add-3)
- ▶ **Input:** 8-bit binary number
- ▶ **Output:** 12-bit BCD (3 digits)
- ▶ **Extraction:** hundreds[11:8], tens[7:4], ones[3:0]

Display Multiplexing:

- ▶ **Refresh Counter:** 18-bit @ 100MHz
- ▶ **Digit Select:** $\text{refresh_counter}[17:16]$
- ▶ **Refresh Rate:** ~381 Hz
- ▶ **Encoding:** Active-low segments & anodes

FSM State Diagram



State Descriptions

IDLE State

- ▶ Reset transaction registers
- ▶ `idle_mode_status = 1`
- ▶ `buy = 1` → PURCHASE

PURCHASE State

- ▶ Accept coins and product selection
- ▶ `go + valid` → DONE
- ▶ `go + invalid` → ERROR

DONE State

- ▶ Deduct stock and calculate change
- ▶ `buy = 1` → PURCHASE
- ▶ Continue with remaining balance

ERROR State

- ▶ Show refund / no stock deduction
- ▶ `buy = 1` → PURCHASE
- ▶ Retry transaction

Debouncing Overview

Counter-Based Debouncing Implementation

- Four independent debouncer instances for coin_btn, pq_btn, buy_btn, go_btn
- No slow clock generation - operates at full 100 MHz system clock
- 10ms debounce threshold (1,000,000 clock cycles @ 100MHz)
- Edge detection provides single-cycle output pulse per button press

Debouncer Architecture

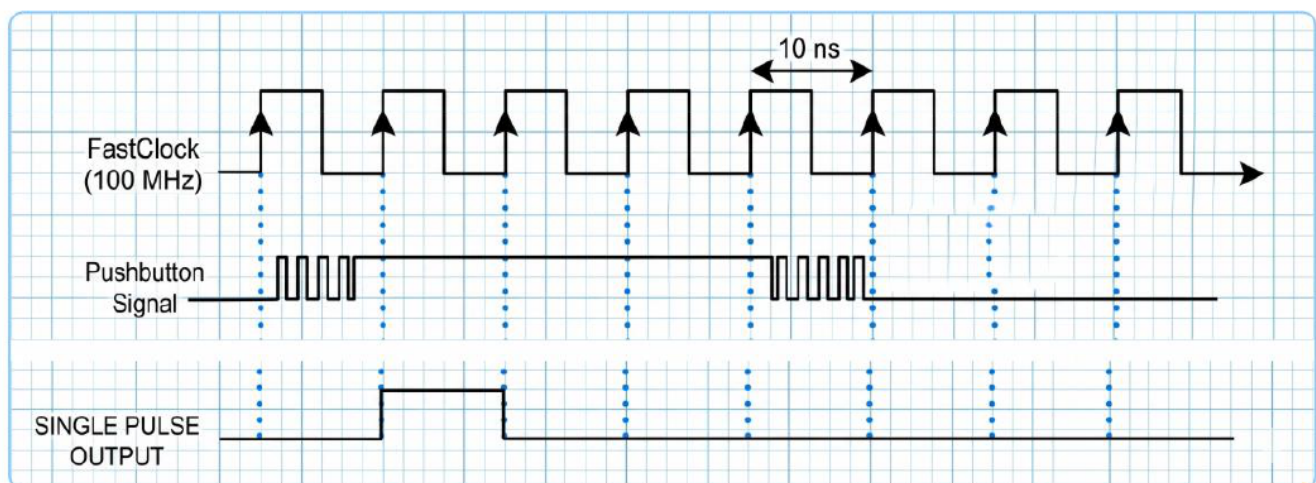
Stage 1: Synchronizer Chain

- **sync_ff1**: First flip-flop (metastability barrier)
- **sync_ff2**: Second flip-flop (double synchronization)
- **sync_ff3**: Third flip-flop (stable synchronized signal)
- **Purpose**: Eliminate metastability from asynchronous button inputs

Stage 2-3: Stability Detection

- **Counter Width**: 20 bits (parameterized)
- **Threshold**: 1,000,000 cycles
- **Logic**: Reset counter on any change, increment when stable
- **Output**: Update button_stable_state when counter $\geq$ threshold

Debouncing Waveform Diagram



Resource Utilization & Performance Analysis

FPGA Resource Utilization (Post-Implementation)

Slice LUTs (Look-Up Tables)

221

Available: 32,600 | Utilization: 0.68%

Slice Registers (Flip-Flops)

184

Available: 65,200 | Utilization: 0.28%

I/O Pins

29

Available: 210 | Utilization: 13.81%

BUFGs (Global Buffers)

1

Available: 32 | Utilization: 3.13%

Timing Analysis Results

Setup Timing		Hold Timing		Pulse Width	
Worst Negative Slack (WNS)	2.832 ns	Worst Hold Slack (WHS)	0.169 ns	Worst Pulse Width Slack	4.500 ns
Total Negative Slack (TNS)	0.000 ns	Total Hold Slack (THS)	0.000 ns	Total Pulse Width Neg Slack	0.000 ns
Failing Endpoints	0	Failing Endpoints	0	Failing Endpoints	0
Total Endpoints	319	Total Endpoints	319	Total Endpoints	185

Power Analysis

Power Report Snapshot

On-Chip Power

Dynamic:	0.015 W	(17%)
Clocks:	0.001 W	(9%)
Signals:	<0.001 W	(3%)
Logic:	<0.001 W	(3%)
I/O:	0.013 W	(85%)
Device Static:	0.072 W	(83%)

Power Summary

- Total On-Chip Power: 0.087 W
- Dynamic Power: 0.015 W (17%)
- Device Static: 0.072 W (83%)
- Junction Temperature: 25.4°C
- Thermal Margin: 59.6°C (12.0 W)
- Effective θ_{JA} : 4.9°C/W
- Confidence Level: Low